Much ado about Toric Codes

October 7, 2022

Contents

1 The stabilizer formalism 1

2 The Stabilizer Code "vibe" 1

3 The Surface Code 3

4 Single Qubit Errors 3

5 Logical Operators 4

1 The stabilizer formalism

2 The Stabilizer Code "vibe"

Let $\hat{X} := \hat{\sigma}_x$, $\hat{Z} := \hat{\sigma}_z$. Errors in QC occur from "random and unpredictable [appearances] of $\hat{X}$, $\hat{Z}$ during the evaluation of a computation. They can be detected by repeated measurements of $\hat{X}$,$\hat{Z}$ but since $\left[\hat{X}, \hat{Z}\right] \neq 0$, such repeated measurements (by causing projections on to the respective eigenstates) destroy the quantum state. For example, a $\hat{Z}$ measurement $M_{\hat{Z}}$ on a qubit with state ψ (randomly) will project that qubit onto $|0\rangle$ or $|1\rangle$ with corresponding (measured) eigenvalues +1 or -1 (respectively), leading to a complete loss of the state's amplitude and phase[1]. A subsequent measurement $M_{\hat{X}}$ of $\hat{X}$ will then project onto the eigenstate $(|0\rangle + |1\rangle)/\sqrt{2}$ or the eigenstate $(|0\rangle - |1\rangle)/\sqrt{2}$ (with corresponding measured eigenvalues $+1$ or -1). Call this the **projective measurement problem**.

The projective measurement problem can be avoided by measuring multiple qubits; consider a two qubit system, with qubits a, b, and the standard $\hat{X}_i, \hat{Z}_i$ where i indicates which qubit the operator will act on. Construct the two-qubit operators $\hat{X}_a\hat{X}_b$ and $\hat{Z}_a\hat{Z}_b$; these operators commute since $\left[\hat{X}_a, \hat{Z}_b\right] = 0$ (since they're applied to different qubits). Thus, these operators can be measured

[1] "Wave function collapse"

$\hat{Z}_a\hat{Z}_b$	$\hat{X}_a\hat{X}_b$	ψ
$+1$	$+1$	$(\|00\rangle + \|11\rangle)/\sqrt{2}$
$+1$	-1	$(\|00\rangle + \|11\rangle)/\sqrt{2}$
-1	$+1$	$(\|01\rangle + \|10\rangle)/\sqrt{2}$
-1	-1	$(\|10\rangle + \|01\rangle)/\sqrt{2}$

Table 1: Eigenstates of the two-qubit operators $\hat{X}_a\hat{X}_b$ and $\hat{Z}_a\hat{Z}_b$.

simultaneously[2]. The possible outcomes of a measurement of both $\hat{X}_a\hat{X}_b$ and $\hat{Z}_a\hat{Z}_b$ appear in table 1.

Consider the state $\psi = (|00\rangle + |11\rangle)/\sqrt{2}$ and notice that an $\hat{X}_b$ error (i.e., $\hat{X}$ error on the second qubit) will transform ψ into $(|01\rangle + |10\rangle)/\sqrt{2}$ *which is also an eigenstate of* $\hat{X}_a\hat{X}_b\hat{Z}_a\hat{Z}_b$ but with eigenvalues $(-1, +1)$. That change - measurements[3] of ψ (by giving $\hat{X}_a\hat{X}_b\hat{Z}_a\hat{Z}_b$) giving eigenvalues $(-1, +1)$ instead of $(+1, +1)$ - reveals the error *without disturbing the wave function* (amplitude and phase) of ψ. Unfortunately, this error cannot be distinguished from a $\hat{X}_a$, which yields the same final state and eigenvalues. The point here is that **while errors can always be detected, they cannot always be uniquely identified using only** $\hat{X}_a\hat{X}_b$ **and** $\hat{Z}_a\hat{Z}_b$. A more complex scheme is necessary.

The operators $\hat{X}_a\hat{X}_b$ and $\hat{Z}_a\hat{Z}_b$ are called **stabilizers**. The core idea of stabilizer codes is repeated measurements of a complete set of commuting stabilizers[4] puts a qubit system into a simultaneous and unique eigenstate of all of them[5]. Since you can measure (the observables corresponding to) the stabilizers without perturbing the system[6], you can catch (some) errors when they occur (evident by transitions from one eigenstate of the stabilizer to another). When stabilizer codes are used, the system is put into a random starting eigenstate of the stabilizers, a so-called **quiescent state** of the system, by simultaneously measuring all stabilizers[7] and then measured repeatedly at regular intervals.

[2] If operators F and G commute, then you can come up with a "complete" set of basis states $|f_n g_n\rangle$ that are eigenstates of both F and G, and thus express any state of the quantum system as

$$|\Psi\rangle = \sum_n \alpha_n |f_n g_n\rangle$$

Measuring F you will get eigenstate f_m with probability $|\alpha_m^2|$ while leaving the system in eigenstate $|f_m g_m\rangle$, and vice-versa for G (i.e., you get the same result either way). Thus, it doesn't matter which order you measure F, G in, you will get a consistent result. This is the significance of commuting operators.

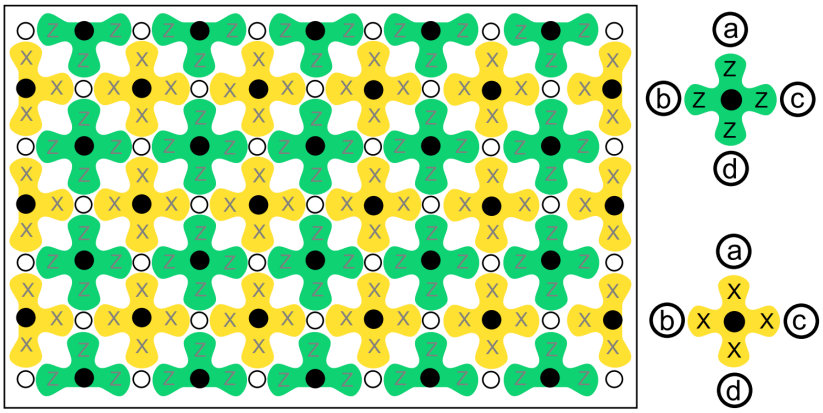
[3] Don't forget: we only know anything about ψ through measurements.

[4] Don't get hung up on complete and commuting - we'll come back to it.

[5] Think about how solving a set of simultaneous equations gives you a unique solution.

[6] That's where the "stable" in stabilizers comes from.

[7] Thus, the system doesn't necessarily start from the groundstate, at the beginning of the quantum computation/quantum circuit.



(a) (b)

Figure 1: (a) A two-dimensional array implementing a surface code. **Data qubits** are open circles (◯), **measurement qubits** are filled circles (●), with measure-$\hat{Z}$ qubits colored green and measure-$\hat{X}$ qubits colored yellow. (b) Neighbor measurement sequence; note that the "zig-zagging" is important.

3 The Surface Code

More complex stabilizer codes exist and are used; the following is a description of such a code, called a **surface code**. A surface code can be implemented using a two-dimensional array of real (physical) qubits, as shown in figure 1a. The physical qubits are one of two types: **data qubits**, within which the computational quantum states are stored, and **measurement qubits**, which are used as stabilizers, but can also alter the state of the data qubits, during, e.g., error correction. There are two types of measurement qubits: **measure-$\hat{Z}$** qubits (measuring the $\hat{Z}$ syndrome) and **measure-$\hat{X}$** qubits (measuring the $\hat{X}$ syndrome). Each data qubit is connected to two of each kind of measurement qubit (two measure-$\hat{Z}$ qubits and two measure-$\hat{X}$ qubits). Conversely, each measurement qubit is connected to four data qubits. **Very important**: measure-$\hat{Z}$ qubit triggers a $\hat{Z}$ measurement on each of its neighboring data qubits ($\hat{Z}_a\hat{Z}_b\hat{Z}_c\hat{Z}_d$ on its a, b, c, d neighbors) and likewise a measure-$\hat{X}$ qubit triggers a $\hat{X}$ measurement on each of its neighboring data qubits (see figure 1b). Therefore, a measure-$\hat{Z}$ measures the stabilizer $\hat{Z}_a\hat{Z}_b\hat{Z}_c\hat{Z}_d$ and measure-$\hat{X}$ measures the stabilizer $\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d$.

4 Single Qubit Errors

Recall that stabilizers codes are initially put into a quiescent state, which thereafter remains unchanged except in the presence of errors. Consider single-qubit $\hat{X}$ spin/bit-flip errors or $\hat{Z}$ phase-flip errors. Such errors show up in the mea-

surement outcomes of the stabilizers (as discussed in section 2). For example

$$\hat{K} := \hat{I}_a + \epsilon \hat{Z}_a$$

affecting data qubit a. This error transform the wavefunction ψ

$$K\psi \to \psi' := \left(\hat{I}_a + \epsilon \hat{Z}_a\right)\psi \equiv \psi + \epsilon \hat{Z}_a \psi$$

What this means is that, upon the next measurement of the neighboring $\hat{Z}_a\hat{Z}_b\hat{Z}_c\hat{Z}_d$ and $\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d$ stabilizers , ψ' will either be projected[8] back to the original ψ (if we're lucky[9]), or it'll be projected to $\hat{Z}_a\psi$ with probability $|\epsilon|^2$. Obviously in the first case, the measurement outcomes of the stabilizers are the same as before the error. On the otherhand, in the second case

$$\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d\left(\hat{Z}_a\psi\right) = -\hat{Z}_a\left(\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d\psi\right) = -X_{abcd}\left(\hat{Z}_a\psi\right)$$

where X_{abcd} is the outcome of the measurement of $\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d$ (i.e., X_{abcd} is a real number). This proves that $\hat{Z}_a\psi$ is an eigenstate of the $\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d$ stabilizer but with the opposite sign of the original ψ. **Very important**: **the same argument applies to the other measure-$\hat{X}$ qubit adjacent to qubit a but such an error has no effect on neighboring measure-$\hat{Z}$ qubits** (since $\hat{Z}$s commute). Thus, we can detect $\hat{Z}$ errors through sign flips in measure-$\hat{X}$ qubits. **A little thinking should reveal that the same is true for $\hat{X}$ errors**. Correcting such errors could involve simply applying the corresponding operators (correct a $\hat{Z}_a$ phase-flip by applying $\hat{Z}_a$ again since $\hat{Z}_a^2 = \hat{I}_a$) but since this could introduce more errors it's usually handled in other ways[1].

5 Logical Operators

How does one perform quantum logic in the surface code? By which we mean, given we're continually measuring the measurement qubits (stabilizing), how (which) useful operations can we perform on the data qubits without interfering with the stabilizers? Put another way: it may seem like process of measuring the stabilizers over and over locks the quantum system into exactly one logical state by virtue of constraining it to always be in an eigenstate of all of the stabilizers. That is actually what we implied at the end of section 2[10]. But, in fact, the set of stabilizers on an array such as 1a is actually not always complete and so quantum system can have some "degrees of freedom". What does completeness mean? Completeness means any possible state can be represented uniquely by some subset of states. This is easiest to understand in terms of vector spaces like $\mathbb{R}^3$; the set of unit vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} \subset \mathbb{R}^3$ is **complete** for $\mathbb{R}^3$ because any vector ψ can be represented as the linear combination

$$\psi \equiv \alpha\mathbf{e}_1 + \beta\mathbf{e}_2 + \gamma\mathbf{e}_3$$

[8] I.e., measured to be in.

[9] If ϵ is very small, i.e., $|\epsilon|^2 \ll 1$.

[10] "simultaneous and unique eigenstate..."

$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is called a **basis** for $\mathbb{R}^3$. Okay but what happens if we forgo the use of say $\mathbf{e}_3$ in trying to represent vectors in $\mathbb{R}^3$? Well then there are a lot of different vectors ψ, ψ' that will have the same representation in terms $\{\mathbf{e}_1, \mathbf{e}_2\}$ but actually be very different. In fact there's a whole line of them right? For fixed α, β the vectors

$$\psi_k := \alpha\mathbf{e}_1 + \beta\mathbf{e}_2 + k\mathbf{e}_3$$

will have the same representation in terms of $\{\mathbf{e}_1, \mathbf{e}_2\}$ (namely (α, β)) but will actually be different vectors (in $\mathbb{R}^3$). That single k is the single "degree of freedom" due to the incomplete set $\{\mathbf{e}_1, \mathbf{e}_2\}$.

Looking at the array in figure 1a, there are 41 data qubits and 40 measure qubits, so there are 2×41 degrees of freedom in the data qubits and 2×40 constraints from the stabilizer measurements[11]. These 2 extra degrees of freedom ($2 \cdot 41 - 2 \cdot 40 = 2$) means the array can serve as a (single) logical qubit. That is to say, we can perform operations on the array that encode useful information along those two free dimensions without interfering/disturbing the stabilizer measurement scheme/process. But how do we find the operators that will allow us to manipulate the additional degrees of freedom in the array, without affecting the stabilizers? The trick is realizing that stabilizer measurements by adjacent stabilizers commute because they share a pair of data qubits (see figure 2a):

$$\begin{aligned}\left[\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d, \hat{Z}_a\hat{Z}_b\hat{Z}_e\hat{Z}_f\right] &= \left(\hat{X}_a\hat{Z}_a\right)\left(\hat{X}_b\hat{Z}_b\right)\hat{X}_c\hat{X}_d\hat{Z}_e\hat{Z}_f - \\ &\quad \left(\hat{Z}_a\hat{X}_a\right)\left(\hat{Z}_b\hat{X}_b\right)\hat{X}_c\hat{X}_d\hat{Z}_e\hat{Z}_f \\ &= \left(\hat{X}_a\hat{Z}_a\right)\left(\hat{X}_b\hat{Z}_b\right)\hat{X}_c\hat{X}_d\hat{Z}_e\hat{Z}_f - \left(-\hat{Z}_a\hat{X}_a\right)\left(-\hat{Z}_b\hat{X}_b\right)\hat{X}_c\hat{X}_d\hat{Z}_e\hat{Z}_f \quad (1) \\ &= 0\end{aligned}$$

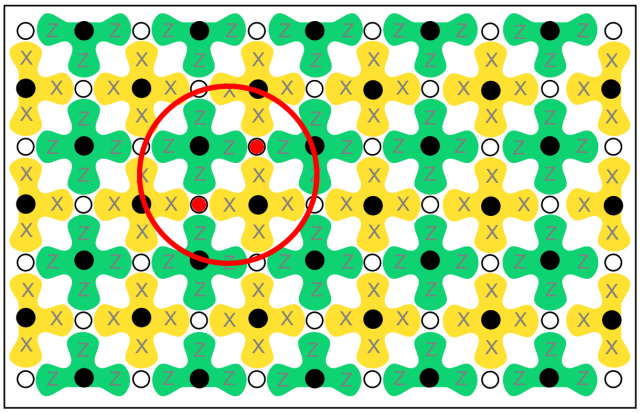
Similarly, products of pairs operators adjacent to the same stabilizer also commute with that stabilizer:

$$\hat{Z}_a\hat{Z}_b\hat{Z}_c\hat{Z}_d\left(\hat{X}_a\hat{X}_b\right)\psi = (-1)^2\hat{X}_a\hat{X}_b\left(\hat{Z}_a\hat{Z}_b\hat{Z}_c\hat{Z}_d\right)\psi = Z_{abcd}\hat{X}_a\hat{X}_b\psi$$

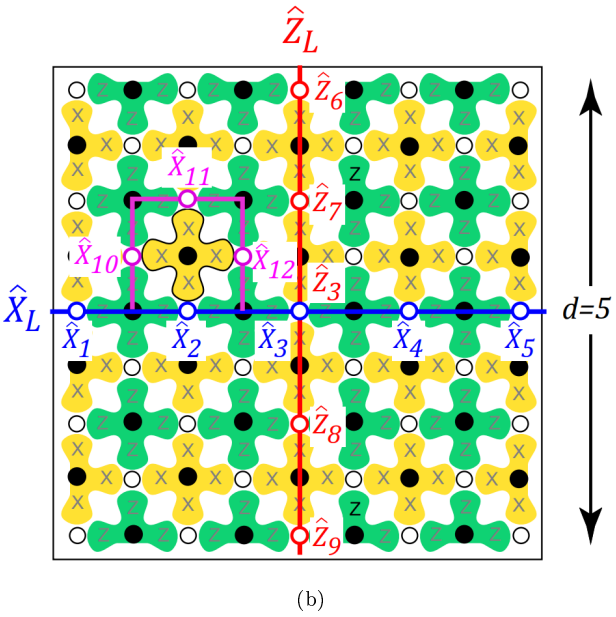
(recall that $Z_{abcd}$is the outcome of the measurement of $\hat{Z}_a\hat{Z}_b\hat{Z}_c\hat{Z}_d$, i.e., a real number). A similar calculation follows for pairs of $\hat{Z}_i$ adjacent to $\hat{X}$ stabilizers. This shows that "product chains" of operators $\hat{Z}_i$ **that stretch across the entire length of the array** commute with all $\hat{X}$ stabilizers and "product chains" of $\hat{X}_i$ commute with all $\hat{Z}$ stabilizers. Picking two such chains arbitrarily $\hat{X}_L$ and $\hat{Z}_L$ we have our logical operators (see figure 2b), in that they each transform a quiescent state ψ into some new state ψ', which while still an eigenstate of all of the stabilizers, cannot be written a product of them[12]. In fact

[11] The state of an n-qubit system lives in $\mathbb{R}^{2^n}$ **but independent measurements of each of the n qubits live in $\mathbb{R}^{2n}$** since for each measurment you get one of two eigenstates.

[12] Each chain is said to "manipulate" 1 degree of freedom by changing $\psi \to \psi' \to \psi''$.



(a)



(b)

Figure 2: (a) Adjacent qubits a, b shared by stabilizers $\hat{X}_a\hat{X}_b\hat{X}_c\hat{X}_d$ and $\hat{Z}_a\hat{Z}_b\hat{Z}_e\hat{Z}_f$. (b) Logical operators $\hat{X}_L$ and $\hat{Z}_L$ as "product chains" as well as a modification of $\hat{X}_L$ to $\hat{X}_L'$ that is equivalent in terms of being a valid logical operator.

the chains need not be "flat" (straight lines); they can be deformed arbitrarily (as long as no crossings are introduced). For example, consider the pink deformation in figure 2b

$$\hat{X}_L^{'}\psi = X_{2,10,11,12}\hat{X}_L\psi = \pm\hat{X}_L\psi = \pm\psi$$

where $X_{2,10,11,12}$ is the outcome of the measurement of stabilizer $\hat{X}_2\hat{X}_{10}\hat{X}_{11}\hat{X}_{12}$.

References

[1] Austin G. Fowler, Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86(3), sep 2012.